

→ Q₁, Q₂, Q₄, Q₅, Q₆, Q₁₀

Q₁ (a)
B=100

$$|V_B = 0|, V_E = 0.7$$

where

$$|V_{EB} = 0.7|, V_E - V_B = 0.7$$

$$V_E = 0.7$$

$$0.7 - V_B = 0.7$$

$$-V_B = 0.7 - 0.7$$

$$|V_B = 0|$$

$$I_1 = \frac{10.7 - 0.7}{10 \times 10^3}$$

$$|I_E = 1 \text{ mA}|$$

$$I_E = (B+1)I_B$$

$$|I_B = 9.9 \times 10^{-6}|$$

$$V_C = ?$$

$$I_C = ?$$

$$I_C = (B+1)I_E$$

$$I_C = \alpha I_E$$

$$I_C = \left(\frac{B}{B+1}\right)I_E$$

$$|I_C = 9.9 \times 10^{-4} \text{ A}|$$

where

$$V_C \text{ or } V_2 = -10.7 + I_2 \times 10 \times 10^3$$

$$V_C = -0.8 \text{ V}$$

$$V_{BC} < -0.4$$

$$V_{BC} > -0.4$$

$$V_B - V_C > -0.4$$

$$V_B - (-0.8) > -0.4$$

$$0.8 > -0.4$$

Active Mode

Active Mode B=100

$$(b) V_{EB} = 0.7$$

where

$$|V_B = -2.7|$$

$$V_E - V_B = 0.7$$

$$V_E = 0.7 + V_B$$

$$|V_E = -2 \text{ V}| \checkmark$$

$$\therefore I_E = \frac{12 - (-2)}{B \cdot 5 \cdot 10^3}$$

$$B = 5 \cdot 10^3$$

$$|I_E = 2.5 \text{ mA}| \checkmark$$

$$\therefore I_E = (B+1)I_B$$

$$|I_B = 2.47 \times 10^{-5} \text{ A}| \checkmark$$

$$\therefore I_E = I_B + I_C \text{ or } I_C = B I_B$$

$$\Rightarrow |I_C = 2.47 \text{ mA}| \checkmark$$

$$V_C = -10 + I_C \times 2.4 \times 10^3$$

$$|V_C = -4.072 \text{ V}| \checkmark$$

operating mode

$$V_{BC} > -0.4$$

$$-2.7 - (-4.072) > -0.4$$

$$1.372 > -0.4$$

g)

$$V_C = 0 \quad B = 100$$

$$V_{EB} = 0.7V$$

$$I_C = \frac{0 - (-10)}{10 \times 10^3}$$

$$I_C = 1mA$$

$$I_C = \beta I_B$$

$$I_B = 1 \times 10^{-5} A$$

$$I_E = (\beta + 1) I_B$$

$$I_E = 1.01 \times 10^{-3} A$$

$$I_C = \alpha I_E$$

$$I_C = 1.01mA$$

$$I_E = I_B + I_C$$

where

$$V_B = I - I_B R_B$$

$$V_B = 0.85V$$

where

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$V_E = 1.55V$$

$$V_C = -10 + I_C R_C$$

$$V_C = 0$$

$$V_{BC} > -0.4$$

$$0.85 - 0 > -0.4$$

$$0.85 > -0.4$$

Active Mode

d

$$V_{BE} = 0.7V$$

$$\beta = 100$$

Pg 2

Loop equation:

$$V_{BE} + I_E R_E - 10 = 0$$

$$0.7 + (I_E \text{ or } I_E) \times 5 \times 10^3 - 10 = 0$$

$$I_E (5 \times 10^3) = 9.3$$

$$I_E = \frac{9.3}{5 \times 10^3}$$

$$\rightarrow I_E = 1.86 \times 10^{-3} A$$

$$I_C = ?$$

$$I_E = (\beta + 1) I_B$$

$$I_B = 1.84 \times 10^{-5} A$$

$$\rightarrow I_C = \beta I_B$$

$$I_C = 1.84 mA$$

Now

$$V_C = V_7$$

$$V_C = 10 - I_C R_C$$

$$V_C = -17.6$$

Where

$$V_E = -10 + I_E \times 5 \times 10^3$$

$$V_E = -0.7$$

$$V_{CB} = -0.4$$

$$-17.6$$

where

$$V_{BE} = 0.7$$

$$V_B - V_E = 0.7$$

$$V_B = 0$$

Q₂
 a) $V_E = 5V$
 $V_C = 2V$

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$V_B = 4.3V$$

where $I_C = \frac{2-0}{1 \times 10^3}$

$$\Rightarrow I_C = 2mA$$

$$I_B = \frac{4.3}{200 \times 10^3} \Rightarrow I_E = I_B + I_C$$

$$I_E = 2.02 \times 10^{-3}$$

$$I_B = 2.15 \times 10^{-5} A$$

where

$$I_C = \beta I_B$$

$$\beta = \frac{I_C}{I_B}$$

$$\beta = 93.02$$

or $I_C = \alpha I_E$
 $\frac{I_E}{I_C} = \alpha$

$$1.01 = \alpha$$

$$1.01 = \frac{\beta}{\beta+1} \Rightarrow (1.01)(\beta+1) = \beta$$

$$I_E = (\beta+1) I_B$$

$$\Rightarrow \frac{I_E}{I_B} - 1 = \beta \Rightarrow \beta = 92.95$$

$$1.01\beta + 1.01 = \beta$$

$$1.01\beta - \beta = -1.01$$

$$(1.01 - 1)\beta = -1.01$$

$$I_C = \alpha I_E$$

$$\frac{I_C}{I_E} = \alpha$$

Q2(a)

Pg 3

$$I_A = 6 \times \frac{3V}{1000V}$$

c) $V_E = 7V$ ✓
 $V_B = 6.3$

$$I_E = \frac{10 - 7}{1 \times 10^3}$$

$$I_E = 3mA$$

as $V_B = I_B R_B$

$$I_B = \frac{V_B}{R_B}$$

$$I_B = \frac{6.3}{100 \times 10^3}$$

$$I_B = 6.3 \times 10^{-5}$$

$$I_E = I_B + I_C$$

$$\Rightarrow I_C = 3mA \text{ or } 2.937mA$$

$$V_C = 2.937$$

$$0.99 = \alpha$$

$$0.99 = \frac{B}{(B+1)}$$

$$0.99 \times (B+1) = B$$

$$0.99B + 0.99 = B$$

$$0.99B - B = -0.99$$

$$-0.01B = -0.99$$

$$(0.99 - 1) B = -0.99$$

$$B = 99$$

b) $V_E = 5V$
 $V_C = 2.3V$

as, $V_{EB} = 0.7$

$$V_E = 0.7 + V_B$$

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$V_B = 5 - 0.7$$

$$V_B = 4.3V$$
 ✓

$$V_B = 4.3V$$

$$I_B = \frac{4.3}{230}$$

$$I_C = \frac{2.3 - 0}{230}$$

$$I_C = 0.01A$$

$$I_B = \frac{4.3 - 2.3}{20 \times 10^3}$$

$$I_B = 1 \times 10^{-4}A$$

$$I_E = I_B + I_C$$

$$I_E = 0.0101A$$

$$\beta = \frac{I_C}{I_B}$$

$$\frac{I_E}{I_B} - 1 = \beta$$

$$\beta = 100$$

$$\beta = 100$$

$$V_E = 0.7$$

$$V_B = 6.3V$$

$$I_E = \frac{10 - 0.7}{1 \times 10^3}$$

$$I_E = 3mA$$

where:-

$$I_C = \frac{V_C - 0}{1 \times 10^3} = \frac{V_C}{1 \times 10^3}$$

$$I_B = \frac{6.3 - V_C}{100 \times 10^3}$$

$$I_B = \frac{6.3}{100 \times 10^3} - \frac{V_C}{100 \times 10^3}$$

$$\text{as } I_E = I_B + I_C$$

$$3 \times 10^{-3} = \frac{6.3}{100 \times 10^3} - \frac{V_C}{100 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$3 \times 10^{-3} - \frac{6.3}{100 \times 10^3} = -\frac{V_C}{100 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$2.935 \times 10^{-3} = V_C \left(\frac{-1}{100 \times 10^3} + \frac{1}{1 \times 10^3} \right)$$

$$2.935 \times 10^{-3} = V_C (9.9 \times 10^{-4})$$

$$V_C = 2.964V$$

c(Q2) Pj(4)

$$I_B = \frac{6.3 - 2.964}{100 \times 10^3}$$

$$I_B = 3.336 \times 10^{-5}A$$

where

$$I_C = \frac{V_C - 0}{1 \times 10^3}$$

$$I_C = 2.964 \times 10^{-3}$$

$$I_E = I_B + I_C$$

$$\Rightarrow I_E = 2.99 \times 10^{-3}A$$

$$\beta = \frac{I_C}{I_B}$$

$$\beta = 88.84$$

and

$$\beta = \frac{I_C}{I_B} - 1$$

$$\beta = 88.92$$

Q4

Saturation Mode
PNP

$$\rightarrow V_{EB} = 0.7$$
$$\rightarrow V_{EC} = 0.2$$

$$V_E - V_B = 0.7$$

$$V_E - V_C = 0.2$$

$$\rightarrow V_E - 0.7 = V_B$$

$$\rightarrow V_E - 0.2 = V_C$$

So,

$$I_C = 5 - (V_E - 0.2)$$

$$V_B = 5 - I_B R_B$$

$$V_E = 0 + I_E R_E$$

$$\rightarrow I_C = \frac{5 - V_C}{10 \times 10^3}$$

$$\rightarrow I_E = \frac{V_E - 0}{1 \times 10^3}$$

$$I_E = \frac{V_E}{1 \times 10^3}$$

$$\rightarrow I_B = \frac{5 - V_B}{100 \times 10^3}$$

Q4:- NPN

$$\rightarrow V_{BE} = 0.7$$

$$\rightarrow V_{CE} = 0.2$$

$$\rightarrow I_C = \frac{5 - V_C}{10 \times 10^3}$$

$$\rightarrow I_E = \frac{V_E}{1 \times 10^3}$$

$$\rightarrow I_B = \frac{5 - V_B}{20 \times 10^3}$$

$$V_B - V_E = 0.7$$

$$V_B = 0.7 + V_E$$

and;

$$V_{CE} = 0.2$$

$$V_C - V_E = 0.2$$

$$V_C = 0.2 + V_E$$

So;

$$\rightarrow I_C = \frac{5 - (0.2 + V_E)}{10 \times 10^3}$$

and

$$\rightarrow I_B = \frac{5 - (0.7 + V_E)}{20 \times 10^3}$$

$$\rightarrow I_C = \dots$$

Q4 // Continuation // Pg (5)

$$I_E = I_B + I_C$$

$$\frac{V_E}{1 \times 10^3} = \frac{5 - 0.7 - V_E}{20 \times 10^3} + \frac{5 - 0.2 - V_E}{10 \times 10^3}$$

$$= \frac{4.3 - V_E}{20 \times 10^3} + \frac{4.8 - V_E}{10 \times 10^3}$$

$$\frac{V_E}{1 \times 10^3} = \frac{4.3}{20 \times 10^3} - \frac{V_E}{20 \times 10^3} + \frac{4.8}{10 \times 10^3} - \frac{V_E}{10 \times 10^3}$$

$$\frac{V_E}{1 \times 10^3} + \frac{V_E}{20 \times 10^3} + \frac{V_E}{10 \times 10^3} = \frac{4.3}{20 \times 10^3} + \frac{4.8}{10 \times 10^3}$$

$$V_E \left(\frac{1}{1 \times 10^3} + \frac{1}{20 \times 10^3} + \frac{1}{10 \times 10^3} \right) = 6.95 \times 10^{-4}$$

$$V_E (1.15 \times 10^{-3}) = 6.95 \times 10^{-4}$$

$$V_E = \frac{6.95 \times 10^{-4}}{1.15 \times 10^{-3}}$$

$$\boxed{V_E = 0.60} \leftarrow$$

as $V_{CE} = 0.2$

$$V_C = 0.2 + V_E$$

$$\boxed{V_C = 0.8}$$

and

$$V_{BE} = 0.7$$

$$V_B = 0.7 + V_E = 1.3$$

and

$$I_E = \frac{V_E}{1 \times 10^3}$$

$$\therefore I_E = 6 \times 10^{-4}$$

$$\therefore I_C = 4.2 \times 10^{-4}$$

$$\therefore I_B = 1.85 \times 10^{-4}$$

verify

$$I_E = I_B + I_C$$

Mode :

$$V_{CB} < -0.4$$

$$0.8 - 1.3 < -0.4$$

$$\boxed{-0.5 < -0.4}$$

Beta forced

$$\beta_f = \frac{I_c(\text{sat})}{I_B}$$

$$\boxed{\beta_f = 2.27}$$

Now calculate currents for this beta:

$$\begin{aligned} \rightarrow I_c &= \beta I_B \\ \rightarrow I_E &= (\beta + 1) I_B \checkmark \\ \rightarrow I_c &= \alpha I_E \end{aligned}$$

→ how to find approx values of collector voltage

P

$$I_D = 5 \text{ mA}$$

$$\frac{5 - V_B}{100 \text{ K}}$$

Q4 (b) PNP

$$\rightarrow \begin{cases} V_{EC} = 0.2 \\ V_{EB} = 0.7 \end{cases}$$

Currents

$$I_E = \frac{5 - V_E}{1 \times 10^3}$$

$$I_C = \frac{V_C - 0}{1 \times 10^3} = \frac{V_C}{1 \times 10^3}$$

$$I_B = \frac{V_B + 5}{10 \times 10^3}$$

as $V_E - V_C = 0.2$

$$V_E = 0.2 + V_C$$

and

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$(0.2 + V_C) - V_B = 0.7$$

$$V_B = 0.2 + V_C - 0.7$$

$$V_B = V_C - 0.5$$

So, $I_E = \frac{5 - (0.2 + V_C)}{1 \times 10^3}$

$$I_E = \frac{4.8 - V_C}{1 \times 10^3}$$

and

$$I_B = \frac{V_C - 0.5 + 5}{10 \times 10^3}$$

$$I_B = \frac{V_C + 4.5}{10 \times 10^3}$$

as;

$$I_E = I_B + I_C$$

$$\frac{4.8 - V_C}{1 \times 10^3} = \frac{V_C + 4.5}{10 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$\frac{4.8}{1 \times 10^3} - \frac{V_C}{1 \times 10^3} = \frac{V_C}{10 \times 10^3} + \frac{4.5}{10 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$\frac{4.8}{1 \times 10^3} - \frac{4.5}{10 \times 10^3} = \frac{V_C}{1 \times 10^3} + \frac{V_C}{1 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$4.35 \times 10^{-3} = V_C (2.1 \times 10^{-3})$$

$$V_C = 2.07$$

and

$$V_E - V_C = 0.2$$

$$V_E = 0.2 + 2.07 = 2.27V$$

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$V_E - 0.7 = V_B$$

$$2.27 - 0.7 = V_B$$

$$V_B = 1.57$$

Pg (6)
PNP

(1) 4 (b)

$$\rightarrow \begin{cases} V_{EC} = 0.2 \\ V_{EB} = 0.7 \end{cases}$$

currents

$$\rightarrow I_E = \frac{5 - V_E}{1 \times 10^3}$$

$$\rightarrow I_C = \frac{V_C - 0}{1 \times 10^3} = \frac{V_C}{1 \times 10^3}$$

$$\rightarrow I_B = \frac{V_B + 5}{10 \times 10^3}$$

as $V_E - V_C = 0.2$

$$\rightarrow V_E = 0.2 + V_C \checkmark$$

and

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$(0.2 + V_C) - V_B = 0.7$$

$$V_B = 0.2 + V_C - 0.7$$

$$\boxed{V_B = V_C - 0.5} \checkmark \checkmark$$

So; $I_E = \frac{5 - (0.2 + V_C)}{1 \times 10^3}$

$$I_E = \frac{4.8 - V_C}{1 \times 10^3}$$

and

$$I_B = \frac{V_C - 0.5 + 5}{10 \times 10^3}$$

$$I_B = \frac{V_C + 4.5}{10 \times 10^3}$$

as;

$$I_E = I_B + I_C$$

$$\frac{4.8 - V_C}{1 \times 10^3} = \frac{V_C + 4.5}{10 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$\frac{4.8}{1 \times 10^3} - \frac{V_C}{1 \times 10^3} = \frac{V_C}{10 \times 10^3} + \frac{4.5}{10 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$\frac{4.8}{1 \times 10^3} - \frac{4.5}{10 \times 10^3} = \frac{V_C}{1 \times 10^3} + \frac{V_C}{1 \times 10^3} + \frac{V_C}{1 \times 10^3}$$

$$4.35 \times 10^{-3} = V_C (2.1 \times 10^{-3})$$

$$\boxed{V_C = 2.07}$$

and

$$V_E - V_C = 0.2$$

$$V_E = 0.2 + 2.07 = 2.27V$$

$$V_{EB} = 0.7$$

$$V_E - 0.7 = V_B$$

$$V_E - 0.2 = V_B$$

$$\boxed{2.57 = V_B} \quad \boxed{V_B = 1.57}$$

$$I_C = R$$

and currents

$$I_E = \frac{5 - V_E}{1 \times 10^3}$$

$$I_E = 2.73 \text{ mA}$$

$$I_C = \frac{V_C}{1 \times 10^3}$$

$$I_C = 2.07 \text{ mA}$$

$$I_B = \frac{V_{B+5}}{10 \times 10^3}$$

$$I_B = 6.57 \times 10^{-4} \text{ A}$$

$$I_E = I_B + I_C$$

$$2.73 \times 10^{-3} = 6.57 \times 10^{-4} + 2.07 \times 10^{-3}$$

$$2.73 \times 10^{-3} = 2.727 \times 10^{-3} \quad \checkmark$$

operating Mode

$$V_{BC} > -0.4$$

$$1.57 - 2.07 > -0.4$$

$$-0.5 > -0.4 \quad \checkmark$$

Beta forced

$$B_f = \frac{I_C(\text{sat})}{I_B} = 3.15$$

Now

$$I_C = B_f \times I_B$$

$$I_C = 2.07 \text{ mA}$$

$$I_C = \alpha I_E$$

$$I_C = 2.07 \text{ mA}$$

$$I_E = (B_f + 1) \times I_B$$

$$I_E = 2.727 \text{ mA}$$

~~✓~~ ✓

how to
find
approximate
values.

Q5) $P_{g(7)}$ PNP $\rightarrow$ (how to find R_E value for Active Mode) $\rightarrow$?

$$B=50$$

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

Loop equation :-

$$-10 + I_E R_E + V_{EB} + I_B R_B + 0 = 0$$

$$-10 + I_E R_E + 0.7 + I_B R_B = 0$$

$$I_E R_E + I_B R_B = 9.3 \rightarrow (1)$$

where $I_E = (B+1) I_B$

$$I_B = \left(\frac{I_E}{B+1} \right) \text{ Put in (1)}$$

$$I_E R_E + \left(\frac{I_E}{B+1} \right) R_B = 9.3$$

$$I_E \left(R_E + \frac{R_B}{B+1} \right) = 9.3$$

$$I_E (5647.058) = 9.3$$

$$\Rightarrow I_E = 1.6468 \text{ mA}$$

Similarly;

$$I_B = \left(\frac{I_E}{B+1} \right)$$

$$\Rightarrow I_B = 3.22 \times 10^{-5} \text{ A}$$

or
 $I_E = (B+1) I_B$
 eq (1) becomes

~~$$(B+1) I_B R_E + I_B R_B = 9.3$$~~

$$(B+1) I_B R_E + I_B R_B = 9.3$$

$$I_B ((B+1) R_E + R_B) = 9.3$$

$$I_B (288000) = 9.3$$

$$I_B = 3.22 \times 10^{-5} \text{ A}$$

Similarly

$$I_E = (50+1) \times I_B$$

$$I_E = 1.64 \times 10^{-3}$$

where

$$I_C = B \times I_B$$

$$I_C = 1.61 \times 10^{-3} \text{ A}$$

Where

$$V_E = 10 - I_E R_E$$

$$V_C = -10 + I_C R_C$$

$$V_B = I_B R_B \quad (\text{as } I_B = \frac{V_B - 0}{R_B})$$

$$\therefore V_B = \frac{V_B}{R_B}$$

a)

$$\Rightarrow V_E = 1.766$$

$$\Rightarrow V_C = -4.687$$

$$\Rightarrow V_B = 1.0626$$

operating Mode

$$V_{BC} > -0.4$$

$$1.0626 - (-4.687) > -0.4$$

$$5.7496 > -0.4$$

Active Mode

b) Active Mode

$$c) R_C = 33K\Omega$$

$$\text{then } V_C = -10 + I_C R_C$$

$$V_C = -10 + (1.61 \times 10^{-3}) (33 \times 10^3)$$

$$V_C = 43.13$$

operating Mode

$$V_{BC} > -0.4$$

$$V_B - V_C > -0.4$$

$$1.0626 - 43.13 > -0.4$$

$$-42.0674 > -0.4$$

The BJT will not remain in Active Mode

$$\text{if } R_C = 33K\Omega$$

as

$$d) V_C = -10 + I_C R_C$$

$$-4.687 + 10 = I_C R_C$$

$$R_C = \frac{-4.687 + 10}{I_C}$$

$$R_C = 3300$$

$$R_C = 3.3K\Omega$$

Q 6 PNP $P_g(8)$

$$\rightarrow V_E = 5V \checkmark$$

$$\rightarrow V_C = 3V \checkmark$$

$$\beta = 50$$

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$V_E - 0.7 = V_B$$

$$V_B = 4.3 \checkmark$$

Currents :-

$$I_B = \frac{V_B}{100 \times 10^3}$$

$$I_B = 4.3 \times 10^{-5} A \checkmark$$

$$I_C = \frac{V_C - 0}{R_C} \checkmark$$

$$\rightarrow I_C = \beta I_B$$

$$I_C = 2.15 \times 10^{-3} \checkmark$$

Now finding value of R_C

$$\text{as } I_C = \frac{V_C}{R_C}$$

$$R_C = \frac{V_C}{I_C}$$

$$R_C = 1395.34 \Omega$$

$$I_C = 2.15 \times 10^{-3}$$

other parameter

$$I_E = (\beta + 1) \times I_B$$

$$I_E = 2.199 \text{ mA} \checkmark$$

if $\beta = 100$ on next Page

Operating Mode

$$V_{BC} > -0.4$$

$$4.3 - 3 > -0.4$$

Active Mode

Now Verifying that

$$\text{for } R_C = 1395.34, V_C = 3V$$

$$V_C = 0 + I_C R_C$$

$$V_C = 2.15 \times 10^{-3} \times 1395.34$$

$$V_C = 2.99 \approx 3V$$

$$\beta = 100$$

$$V_E = 5V$$

$$V_B = V_E - 0.7$$

$$V_B = 4.3$$

$$V_C = +3V$$

Currents:

$$I_B = \frac{V_B}{100k\Omega}$$

$$\rightarrow I_B = 4.3 \times 10^{-5} A$$

$$\rightarrow I_C = \beta \times I_B$$

$$\boxed{I_C = 4.3mA}$$

$$\rightarrow I_E = ?$$

$$I_C = \alpha I_E$$

$$\boxed{I_E = 4.343mA}$$

or

$$I_E = I_B + I_C$$

$$\boxed{I_E = 4.343mA}$$

Value of R_C :

$$I_C = \frac{V_C}{R_C}$$

$$R_C = \frac{V_C}{I_C}$$

$$R_C = 697.67$$

Verifying at $R_C = 697.67$

$$V_C = +3$$

$$\text{as } V_C = 0 + I_C R_C$$

$$\checkmark \boxed{V_C = 2.99}$$

→ Currents are changed at $\beta = 100$ while operating Mode:

$$V_{BC} > -0.4$$

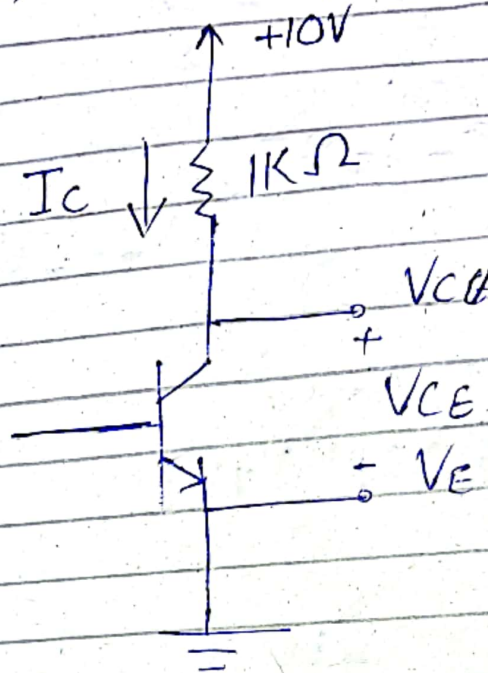
$$4.3 - 3 > -0.4$$

Active Mode.

Q:-

Common emitter configuration:-

$$V_{CC} = +10V, \quad R_C = 1K\Omega$$



1) $I_C = 1mA$

$$V_C = 10 - I_C R_C$$

$$V_C = 9$$

as $V_E = 0$

$$V_{CE} = V_C - V_E$$

$$V_{CE} = 9V$$

$$A_v = - \frac{(V_{CC} - V_{CE})}{V_T} \Rightarrow |A_v| = -40V$$

•) Positive signal swing

$$V_{CC} - V_{CE} = 10 - 9$$

$$V_{CC} - V_{CE} = 1V$$

•) negative signal swing

$$V_{CE(EO)} - V_{CE}''$$

$$0.5 - 9 = -8.5$$

•) $I_C = 5mA$

$$V_C = 10 - I_C R_C$$

$$V_C = 10 - (5 \times 10^{-3})(1 \times 10^3)$$

$$V_C = 5V$$

$$V_{CE} = V_C - V_E$$

$$V_{CE} = 5V$$

$$V_E = 0V$$

where

$$A_v = - \frac{(V_{CC} - V_{CE})}{V_T}$$

$$A_v = - \frac{(10 - 5)}{25 \times 10^{-3}}$$

$$A_v = -200$$

→ Positive and negative signal swing:-

•) +ve signal swing:-

$$V_{CC} - V_{CE} = 10 - 5 = 5V$$

•) -ve signal swing:-

$$V_{CE(sat)} - V_{CE} = 0.3 - 5 = -4.7V$$

•) $I_C = 8mA$

$$V_C = 10 - 8 \times 10^{-3} \times 1 \times 10^3$$
$$V_C = 2V$$

where

$$V_{CE} = V_C - V_E$$
$$V_{CE} = 2V$$

$$A_v = - \frac{(V_{CC} - V_{CE})}{V_T}$$

$$A_v = - \frac{(10 - 2)}{25 \times 10^{-3}}$$

$$A_v = -320$$

•) $I_C = 9mA$

$$V_C = 10 - 9 \times 10^{-3} \times 1 \times 10^3$$
$$V_C = 1V$$

where

$$V_{CE} = 1V$$

Positive and negative signal swing:-

•) +ve signal swing:-

$$V_{CC} - V_{CE}$$
$$5 - 2 = 3V$$

•) -ve signal swing:-

$$V_{CE(sat)} - V_{CE}$$
$$= 0.3 - 2$$
$$= -1.7$$

and

$$A_v = - \frac{V_{CC} - V_{CE}}{V_T}$$

$$A_v = -360$$

1) Positive and negative signal swing:-

→ +ve signal swing:-

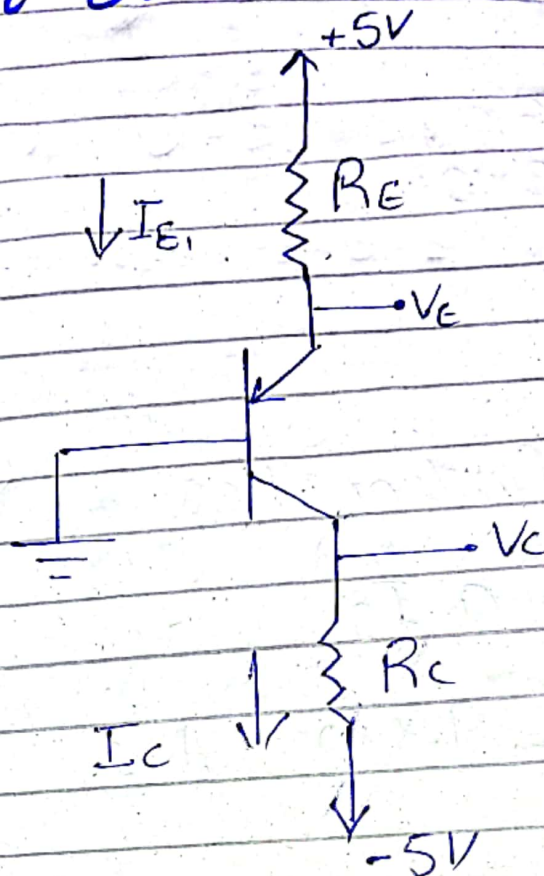
$$\begin{aligned} V_{CC} - V_{CE} \\ 10 - 1 = 9V \end{aligned}$$

2) -ve signal swing:-

$$\begin{aligned} V_{CE(sat)} - V_{CE} &= 0.3 - 1 \\ &= -0.7 \end{aligned}$$

Question 8:-

$$\beta = 100$$



$$|V_C = -1V|, |I_E = 1mA|$$

Assume Active Mode

$$V_{EB} = 0.7$$

$$\text{while } |V_B = 0|$$

$$\rightarrow |V_E = 0.7|$$

$$I_E = 1mA$$

$$\Rightarrow R_E = \frac{5V - V_E}{1 \times 10^{-3}}$$

$$R_E = 4.3K\Omega$$

Verifying I_E -

$$I_E = \frac{5 - 0.7}{4.3 \times 10^3}$$

$$I_E = 1 \text{ mA}$$

→ Similarly finding R_C -

$$I_C = \alpha I_E$$

$$I_C = 9.9 \times 10^{-4} \text{ A}$$

So,

$$R_C = \frac{V_C + 5}{9.9 \times 10^{-4}}$$

$$R_C = \frac{-1 + 5}{9.9 \times 10^{-4}}$$

$$R_C = 4040.40 \Omega$$

Verifying I_C

$$I_C = \frac{V_C + 5}{4040.40}$$

$$I_C = \frac{-1 + 5}{4040.40}$$

$$I_C = 9.9 \times 10^{-4}$$

$$\Rightarrow |V_C = I_C R_C - 5 = -1V| \quad \text{verified}$$

where verifying emitter current :-

$$\text{as } I_C = 9.9 \times 10^{-4}$$

$$I_C = \alpha I_E$$

$$\frac{9.9 \times 10^{-4}}{0.99} = I_E$$

$$I_E = 1.00 \text{ mA}$$

or,

$$I_E = \frac{5 - V_E}{R_E}$$

$$I_E = \frac{5 - 0.7}{4.3 \times 10^3}$$

$$I_E = 1 \text{ mA}$$

Operating Mode :

$$V_{BC} > -0.4$$

$$0 - (-1V) > -0.4$$

$$1V > -0.4$$

$$I_C = 9.9 \times 10^{-4}$$

$$\Rightarrow V_C = I_C R_C - 5 = -1V \quad \text{verified}$$

where verifying emitter current :-

$$\text{as } I_C = 9.9 \times 10^{-4}$$

$$I_C = \alpha I_E$$

$$\frac{9.9 \times 10^{-4}}{0.99} = I_E$$

$$I_E = 1.00 \text{ mA}$$

Or,

$$I_E = \frac{5 - V_E}{R_E}$$

$$I_E = \frac{5 - 0.7}{4.3 \times 10^3}$$

$$I_E = 1 \text{ mA}$$

Operating Mode :

$$V_{BC} > -0.4$$

$$0 - (-1V) > -0.4$$

$$1V > -0.4$$

Now;

$$V_{EB} = 0.64 \text{ at } I_E = 0.1 \text{ mA} \leftarrow$$

$$B = 100$$

$$R_C = ?$$

$$I_C = I_{se} \frac{V_{EB2}}{n V_T}$$

$$n = 1$$

$$I_C = I_{se} \frac{V_{EB2}}{n V_T}$$

Using,

$$V_{EB2} - V_{EB1} = 2.3 n V_T \log_{10} \frac{I_{E2}}{I_{E1}}$$

$$V_{EB2} = V_{EB1} + 2.3 \times 1 \times 25 \times 10^{-3} \log_{10} \frac{0.1 \text{ mA}}{1 \text{ mA}}$$

$$V_{EB2} = 0.64 + 2.3 \times 1 \times 25 \times 10^{-3} \log_{10} \frac{0.1 \text{ mA}}{1 \text{ mA}}$$

$$V_{EB2} = 0.582 \text{ V}$$

Using;

$$I_C = I_{se} \frac{V_{EB2}}{n V_T}$$

$$I_C = 1 \times 10^{-15} \times e^{\frac{0.582}{25 \times 10^{-3} \times 1}}$$

$$I_C = 1.28 \times 10^{-5}$$

finding R_C

$$R_C = \frac{V_C + 5}{1.28 \times 10^{-5}}$$

$$R_C = \frac{-1 + 5}{1.28 \times 10^{-5}}$$

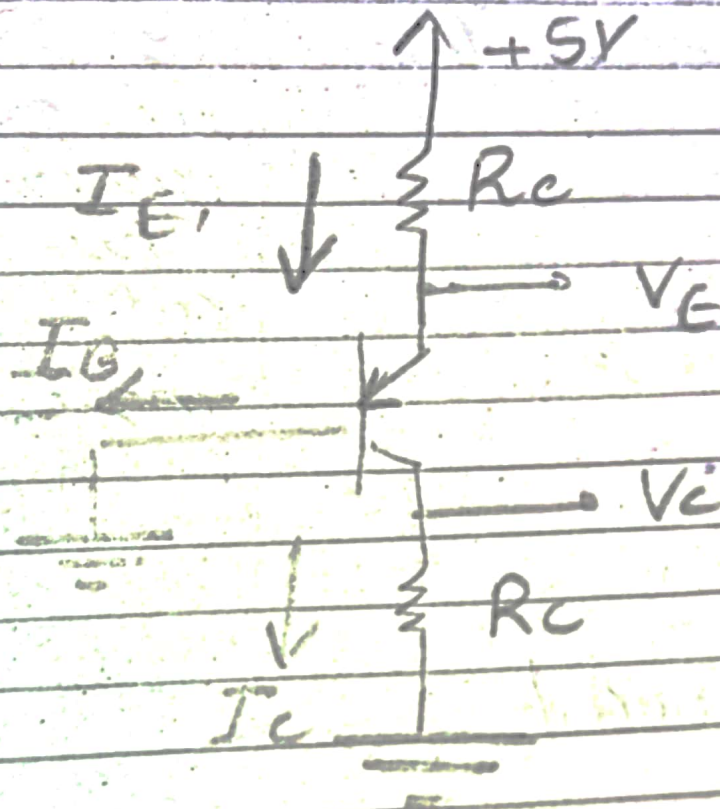
$$R_C = 312500 \Omega$$

Question 8 :-

$$\rightarrow |I_E = 1\text{mA}|, V_C = -1\text{V}$$

$$V_{EB} = 0.64\text{V at } 0.1\text{mA} = I_E$$

$$\beta = 100$$



$$\rightarrow R_E = \frac{5 - V_E}{I_E} \rightarrow \textcircled{1}$$

$$\rightarrow R_C = \frac{V_C + 5}{I_C} \rightarrow \textcircled{2}$$

as

$$V_{BE2} - V_{BE1} = 2.3 n V_T \log_{10} \left(\frac{I_{E2}}{I_{E1}} \right)$$

$$V_{BE2} = V_{BE1} + 2.3 \times 1 \times 25 \times 10^{-3} \log_{10} \left(\frac{0.1 \times 10^{-3}}{1 \times 10^{-3}} \right)$$

$$V_{BE2} = 0.64 - 0.057$$

$$V_{BE2} = 0.58$$

where $V_{B2} = 0$

so

$$\boxed{V_{E2} = 0.58}$$

from (1)

$$R_E = \frac{5 - V_E}{I_E}$$

$$R_E = \frac{5 - 0.58}{1 \times 10^{-3}}$$

$$R_E = 4.4 \times 10^3 \Omega$$

from (2)

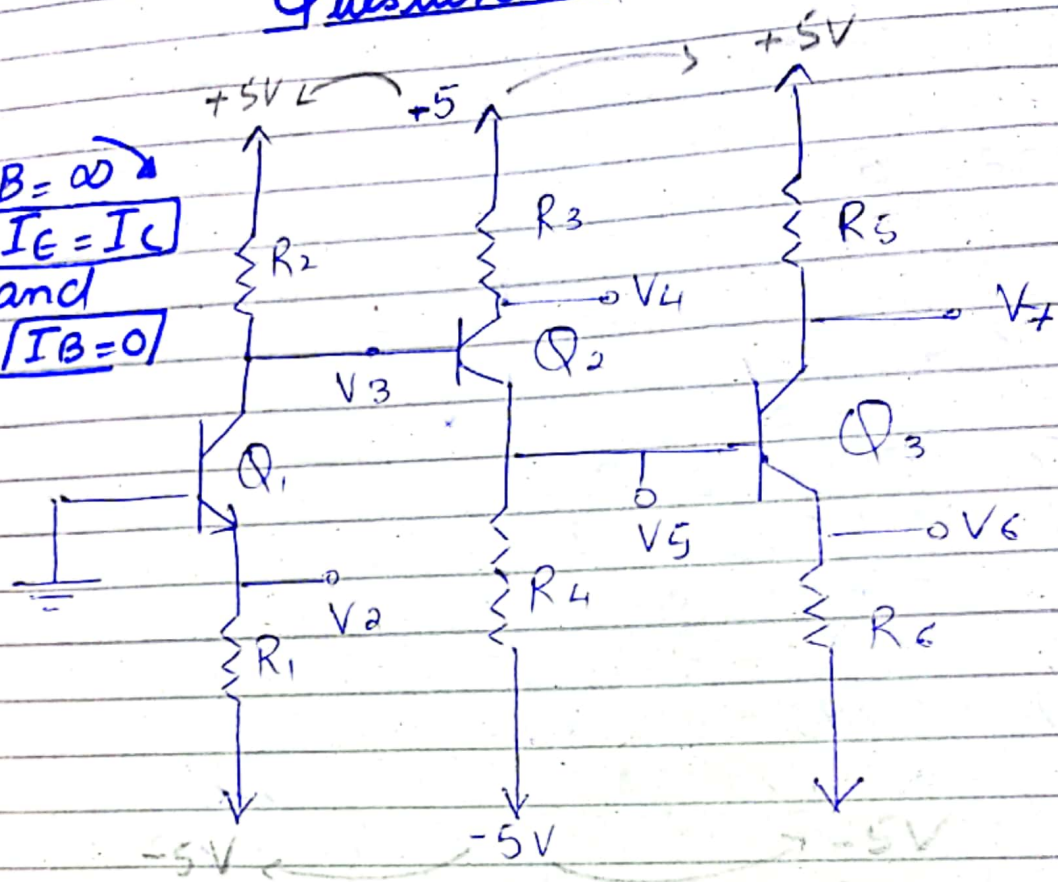
$$R_c = \frac{-1 + 5}{\alpha \times I_c}$$

$$R_c = \frac{4}{0.99 \times 1 \times 10^{-3}}$$

$$[R_c = 4.04 \times 10^3 \Omega]$$

Question #09

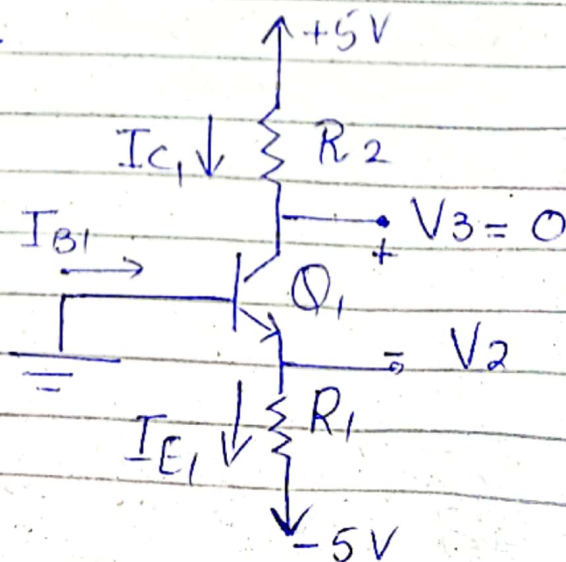
$B = \infty$
 $I_E = I_C$
 and
 $I_B = 0$



$\rightarrow$ Q_1 , Q_2 , and Q_3 biasing currents are 1mA , 1mA , and 2mA respectively

where $V_3 = 0$, $V_5 = -2\text{V}$, $V_7 = 1\text{V}$

Analyzing :-



R₂:-

$$R_2 = \frac{5 - V_3}{1 \times 10^{-3}}$$

$$R_2 = \frac{5}{1 \times 10^{-3}}$$

$$R_2 = 5 \text{ K}\Omega$$

$$I_1 = \frac{5 - V_3}{R_2}$$

$$\Rightarrow \boxed{I_{C1} = \frac{5 - 0}{5 \times 10^3} = 1 \text{ mA}}$$

$$I_E = I_B + I_C$$

R₁:-

where $V_B = 0$

$$V_{BE} = 0.7$$

$$V_E = 0.7$$

$$\boxed{V_2 = 0.7}$$

$$V_2 = V_E$$

$$V_B - V_E = 0.7$$

$$\boxed{V_E = -0.7}$$

$$\text{So, } R_1 = \frac{V_2 + 5}{1 \times 10^{-3}}$$

$$\left(\frac{V_2 - (-5)}{I_1} \right)$$

$$R_1 = \frac{-0.7 + 5}{1 \times 10^{-3}}$$

$$\rightarrow 4.3 \times 10^3 \Omega$$

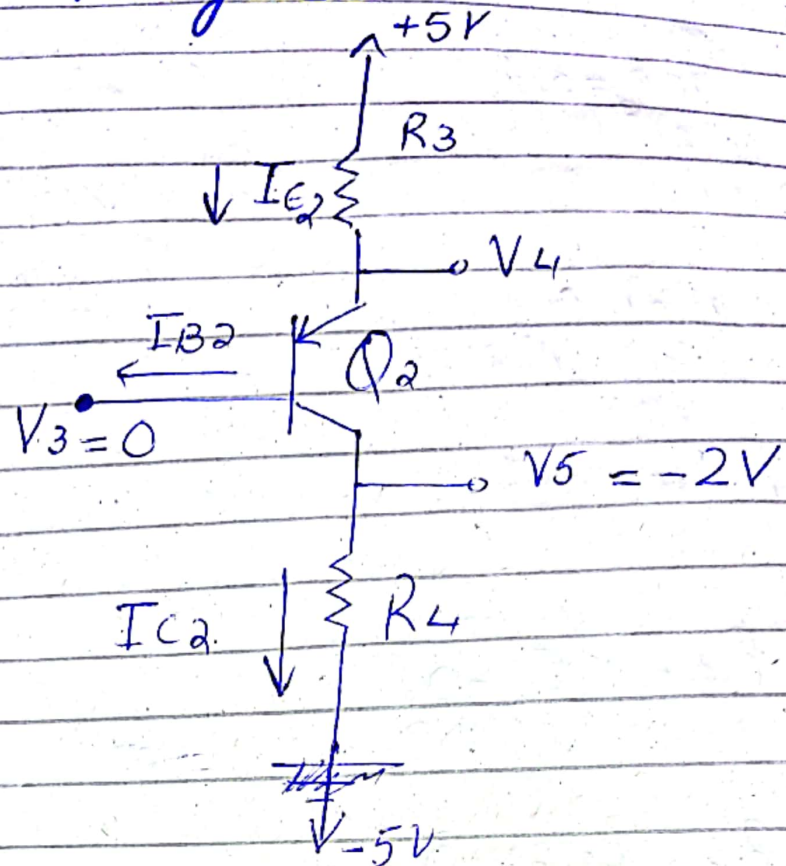
$$R_1 = 4.3 \times 10^3 \Omega$$

So,

$$I_{E1} = \frac{V_2 - (-5)}{R_1}$$

$$\boxed{I_{E1} = \frac{0.7 + 5}{4.3 \times 10^3} = 1 \text{ mA}}$$

Stage 2 Analysis:-



R_4 :-

as biasing current $I_2 = 1\text{mA}$

$$R_4 = \frac{-2 + 5}{1 \times 10^{-3}} \quad \therefore \left(\frac{V_5 - (-5)}{I_2} \right)$$

$$R_4 = 3 \times 10^3 \Omega$$

where

$$I_{C2} = \frac{V_2 - (-5)}{R_4}$$

$$I_{C2} = \frac{-2 + 5}{3 \times 10^3}$$

$$I_{C2} = 3\text{mA}$$

R₃ :-

$$V_{E2} = 0.7$$

$$V_{E2} - V_{B2} = 0.7 \quad \text{where } V_{B2} = 0$$

$$V_{E2} = 0.7$$

$$V_4 = V_{E2}$$

$$\rightarrow \boxed{V_4 = 0.7V}$$

so

$$R_3 = \frac{5 - 0.7}{1 \times 10^{-3}}$$

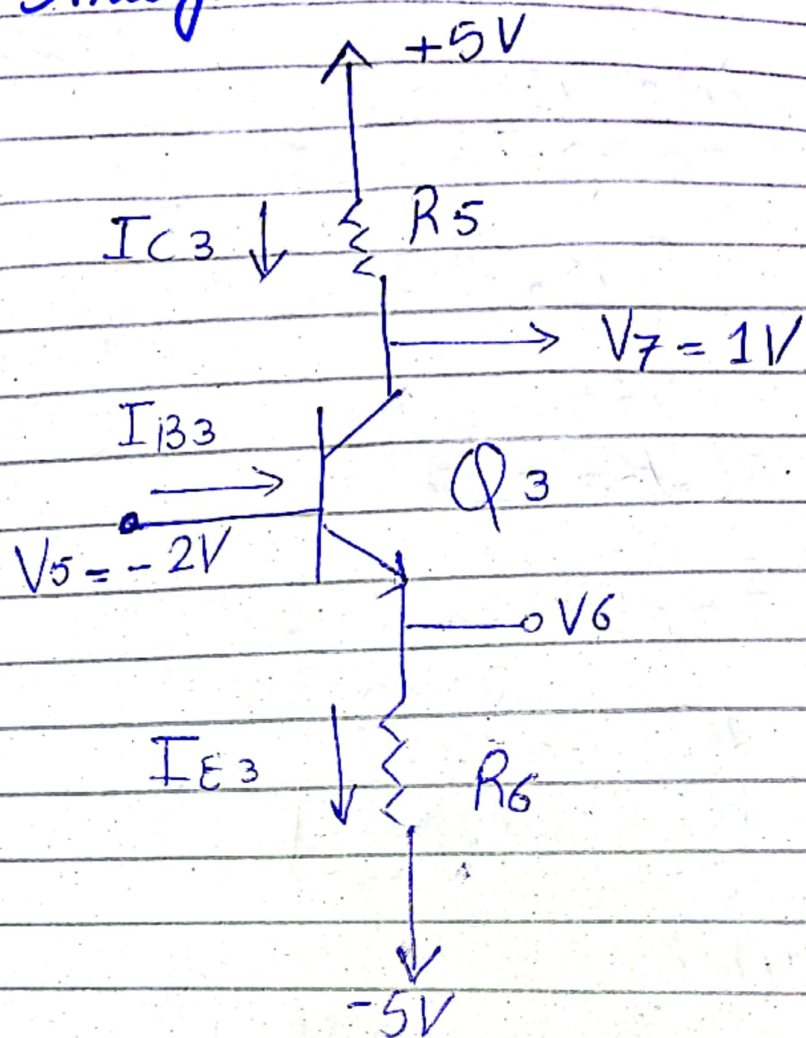
$$R_3 = 4.3 \text{ K}\Omega$$

where,

$$I_{E2} = \frac{5 - 0.7}{4.3 \times 10^3}$$

$$I_{E2} = 1 \text{ mA}$$

Stage 3 Analysis:-



R_5 :-

$$R_5 = \frac{5 - 1}{2 \times 10^{-3}}$$

$$R_5 = 2K\Omega$$

where,

$$I_{C3} = \frac{5 - 1}{2 \times 10^3}$$

$$\rightarrow I_{C3} = 2mA \quad \checkmark$$

R₆ :-

$$V_{BE3} = 0.7$$

$$V_{B3} - V_{E3} = 0.7$$
$$-2 + 0.7 = V_{E3}$$

where

$$V_{B3} = -2$$

$$\boxed{V_{E3} = -1.3 \text{ V}}$$

$$V_6 = V_{E3}$$

$$\boxed{V_6 = -1.3}$$

$$R_6 = \frac{V_6 - (-5)}{2 \times 10^{-3}}$$

$$R_6 = 1850 \Omega$$

where

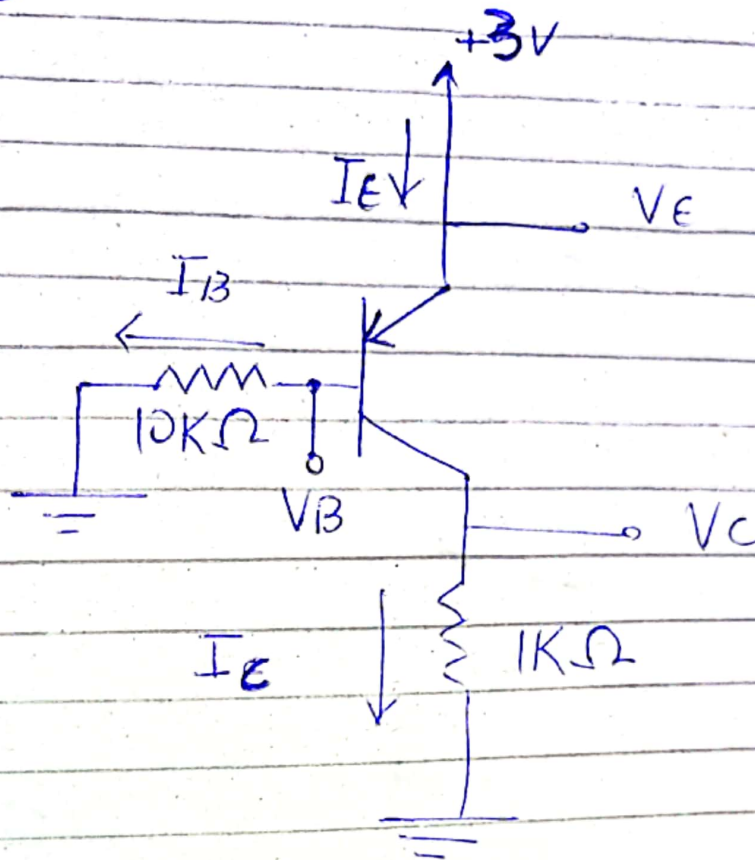
$$I_{E3} = \frac{V_6 - (-5)}{R_6}$$

$$I_{E3} = \frac{-1.3 + 5}{1850}$$

$$\boxed{I_{E3} = 2 \text{ mA}}$$

Q10

$$\beta = 50$$



$$\rightarrow |V_E = 3V|$$

Assuming Mode of Saturation:-

$$V_{EB} = 0.7, \quad V_{EC} = 0.2$$

$$V_E - V_B = 0.7, \quad V_E - V_C = 0.2$$

$$3 - 0.7 = V_B, \quad 3 - 0.2 = V_C$$

$$|V_B = 2.3|$$

$$|V_C = 2.8|$$

where;

$$I_B = \frac{V_B}{10K\Omega}$$

$$I_B = \frac{2.3}{10 \times 10^3}$$

$$I_B = 2.3 \times 10^{-4} \text{ A}$$

Similarly

$$I_C = \frac{V_C}{1 \times 10^3}$$

$$I_C = \frac{2.8}{1 \times 10^3}$$

$$I_C = 2.8 \text{ mA}$$

where;

$$I_E = I_B + I_C$$

$$I_E = 3.03 \times 10^{-3} \text{ A}$$

Operating Mode

$$V_{BC} > -0.4$$

$$2.3 - 2.8 > -0.4$$

$$-0.5 > -0.4$$

$$\text{Beta forced} = \frac{I_C(\text{sat})}{I_B}$$

$$= \frac{2.8 \times 10^{-3}}{2.3 \times 10^{-4}}$$

$$\beta_f = 12.17$$

we can calculate currents with this Beta forced in saturation mode, not with specified one Beta.

(b) $R_B = ?$ EOS

$\Rightarrow$

$$V_{BC} = -0.4$$

(EOS)

where

$$V_{EB} = 0.3$$

$$V_E - V_C = 0.3$$

$$3 - 0.3 = V_C$$

$$V_C = 2.7$$

$$V_B - V_C = -0.4$$

$$V_B = -0.4 + 2.7$$

$$V_B = 2.3$$